

Nguyễn Đức Hữu Thiện

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Skills

Languages: JavaScript, TypeScript, Java, Python

Frontend: Angular, ReactJs, NextJs

Backend: Java Servlet, REST API, MVC

Database: MySQL

Tools: Git, Vercel, Nginx, Directus CMS, Figma

AI & Mathematic: YOLOv8, OpenCV, Genetic Algorithm

Fresher Software Engineer with experience in backend development, data processing, frontend web applications, AI/computer vision, and financial optimization systems. Passionate about backend engineering and data handling, while also expanding into frontend development to build complete end-to-end applications. Currently pursuing a fullstack development path with an interest in prompt engineering and AI-driven systems.

Projects

Thesis Project

May 2025 - Aug 2025

Diverty–Stock Portfolio Optimization Web Application

Backend - (Next.js, TypeScript, REST API, Yahoo Finance API, Genetic Algorithm, Modern Portfolio Theory, Vercel)

- Collaborated in designing application architecture, **frontend-backend workflow**, and state management strategy using **NextJs** and **TypeScript**.
- Developed backend **REST APIs** and integrated **Yahoo Finance** market data for stock searching and historical price analysis.
- Implemented financial analysis modules including return/risk calculation, covariance/correlation matrices, and **Efficient Frontier** generation using **Modern Portfolio Theory**.
- Developed a **Genetic Algorithm-based** optimization system to generate portfolios according to user-selected strategies such as minimum risk, maximum return, target return/risk, and optimized allocation.
- Improved application responsiveness by running computationally intensive calculations **asynchronously** with state-controlled background processing to prevent blocking user interactions.
- Performed **manual testing**, implemented **error handling validation**, and **deployed** the application on Vercel with **automated CI/CD** integration.

Internship

Vietnam German University Landing Page

Frontend - (Angular, TypeScript, Directus CMS, REST API, Nginx, Figma)

Sep 2024 - Feb 2025

- Analyzed client requirements and translated business ideas into functional web interface designs.
- Designed** multiple **UI pages in Figma**, including university landing pages, organization pages, and program information pages.
- Developed **responsive** frontend interfaces using **Angular** and **TypeScript**.
- Built **reusable dynamic UI components** integrated with **Directus CMS** through **REST APIs**.
- Performed **manual testing** and **error handling validation** to improve application reliability and user experience.
- Deployed** the application to the university's private server using **Nginx** and implemented **Git hook-based CI/CD** automation.

University Project

AI-based Mango Ripeness Classification System

Sep 2023 - Jan 2024

AI Developer - (Python, YOLOv8, OpenCV, Computer Vision)

- Collected** and **annotated real-world image datasets** of Vietnamese "Xoài Keo" mangoes for supervised AI training.
- Developed a **computer vision** module using **Python**, **OpenCV** (cv2), and **YOLOv8** to classify mango ripeness states: unripe, ripe, and overripe.
- Trained and fine-tuned a pretrained YOLO model for image classification tasks.
- Developed a web-based application allowing users to upload mango images and receive ripeness predictions.

University Project

Student Information Management System

Oct 2020 - Mar 2021

Backend - (Java Servlet, MySQL, REST API, MVC)

- Developed **backend features** for a student information management application using **Java Servlet** and **MySQL**.
- Designed **relational database** schemas and implemented **CRUD** operations following **REST API** and **MVC** architecture principles.
- Implemented **database querying** and **API response** handling for frontend integration.

Achievements

2017

Vietnam National Champion, World Robot Olympiad. Represented Vietnam at the international competition in Costa Rica, working on embedded systems and robotics programming.

EDUCATION

Computer Science

Vietnamese German University

Sep 2018 - Nov 2025

GPA: 2.3 (German scale) - 7.7 (Vietnamese scale)

LANGUAGE

- Vietnamese: Native
- English: Fluent (IELTS score 6.0 in 2018)
- German: Basic (Completed B1 coursework)